

# Dr. rer. nat. Sebastian Lapuschkin

\* December 16, 1986 in Würzburg

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## Short Bio

Sebastian Lapuschkin heads the Explainable Artificial Intelligence research group at the Fraunhofer Heinrich Hertz Institute (HHI) in Berlin, directing a team of more than 25 researchers. He received his Ph.D. with distinction from the Technische Universität Berlin in 2018 for foundational contributions to Explainable AI (XAI) and interpretable machine learning.

His research investigates the inner workings of modern frontier models: mechanistic interpretability of Transformers and large language models, the attribution of model behavior to training data and internal components, and the use of these insights to debug, control, and compress neural networks. Recent contributions include real-time attribution for LLMs and

Vision Transformers (AttnLRP), attention-head-level analysis of in-context learning, and explanation-guided pruning that removes up to 95% of model parameters without loss of performance.

His foundational methods — among them Layer-wise Relevance Propagation and Concept Relevance Propagation — are widely used across academia and industry. He is a committed advocate of open science, releasing open-source toolkits such as Zennit and Quantus to support reproducible AI research, and is a recipient of the Hugo-Geiger-Prize and the 2020 Pattern Recognition Best Paper Award.

Since 2024 he co-organizes the World Conference on eXplainable Artificial Intelligence.

## Professional Experience

Technological University Dublin

DUBLIN, IRELAND

### External Scholar

2025 -

at the Centre of eXplainable Artificial Intelligence. The Centre is the first of its kind in the Republic of Ireland and it aims to increase further and synergise cross-centres, college and external research collaboration.

Fraunhofer Heinrich-Hertz-Institute

BERLIN, GERMANY

### Contact Person

2025 -

for the Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFRoS) associate partnership of Fraunhofer HHI.

### Ethics Committee Member

2023 -

Founding member of the first ethics committee at Fraunhofer HHI.

### Head of Explainable Artificial Intelligence

2021 -

Research Group Leadership and direction of XAI research & applications.

### Principal Investigator

2019 - 2020

Tenured PostDoctoral researcher in the Machine Learning Group at Fraunhofer HHI.

### Research Associate

2014 - 2018

Founding member of the Machine Learning Group at Fraunhofer HHI.

Technische Universität Berlin

BERLIN, GERMANY

### Research Associate

2013 - 2014

Supervision by Prof. Dr. Klaus-Robert Müller.

### Student Research- & Teaching Assistant

2011 - 2013

Research & Teaching assistant to Prof. Dr. Klaus-Robert Müller.

### Student Teaching Assistant

2009 - 2011

Teaching assistant to Prof. Dr. Marc Alexa, Prof. Dr. Odej Kao and Prof. Dr. Oliver Brock.

## Academic Service

### Explainable and Ethical AI Workshop

**Program Chair** 2026 -  
of the 4th workshop “Explainable and Ethical AI” (XAIE4, 2026), co-located with ICPR 2026 in Lyon, France.

### XAI4Science

**Organizer** 2024 -  
of the 1st workshop “XAI4Science: From Understanding Model Behavior to Discovering New Scientific Knowledge (2025)”, co-located with ICLR 2025 at Singapore EXPO, Singapore.  
TBA: 2nd of the workshop at AAAI 2026, Singapore.

### Multidisciplinary Digital Publishing Institute (MDPI)

**Topic Editor** 2024 - 2026  
for “Opportunities and Challenges in Explainable Artificial Intelligence”.  
Submission pre-screening, review management and decision handling.

### World Conference on eXplainable Artificial Intelligence

**Steering Committee Member** 2024 -  
for the 3rd XAI World Conference (2025) in Istanbul, Turkey,  
for the 4th XAI World Conference (2026) in Fortaleza, Brazil ,  
as well as future undisclosed instances of the conference.  
Panelist, Conference and Special Track co-organization.

**Programme Committee Chair** 2023 - 2024  
for the 2nd XAI World Conference (2024) in Valetta, Malta.  
Conference and Special Track co-organization.

### Reviewer & PC Memberships

**Reviewer, Meta-Reviewer / Area Chair, for selected venues**  
NeurIPS, ICML, ICLR, CVPR, ICIP, ACL, TMLR, PLoS ONE, Nature Machine Intelligence, ...

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## Education

### Technische Universität Berlin

BERLIN, GERMANY  
2018

**PhD in Machine Learning (*summa cum laude*)**  
*Date of oral defense: December 19<sup>th</sup>, 2018.*  
*Dean's signature on Doctorate Certificate dated January 23<sup>rd</sup>, 2019.*

Thesis: “Opening the machine learning black box with Layer-wise Relevance Propagation”  
Supervision headed by Prof. Dr. Klaus-Robert Müller.

**Master of Science in Computer Science** 2013  
Focus on machine learning, computer vision and large scale data analysis.

**Bachelor of Science in Computer Science** 2010  
Focus on algorithms and software development

### Deutschhaus-Gymnasium

WÜRZBURG, GERMANY  
2007

**Abitur** (pre-university secondary education)

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## Teaching

Teaching of and teaching support for 19 university courses since 2009, including

“**Responsible Artificial Intelligence 1**”, Technische Universität Berlin 2023 -  
“(An Introduction to) Explainable AI”, Universitat de Girona / IFRoS 2021 -

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## Talks & Lectures

Over 30 invited talks and individual lectures held since 2017, including

**Explainability in the Era of LLMs: New Challenges and Pathways to Actionable Insights** 2025  
2025 Workshop on Self-Supervised Learning for Signal Decoding, Aalborg, Denmark

**XAI as a Tool Beyond Model Understanding – From Heatmaps to Concepts and XAI Automation** 2024  
CBS CoCoNUT | Max Planck Institute for Human Cognitive and Brain Sciences

**Explaining AI with Concept Relevance Propagation** 2023  
4<sup>th</sup> Japanese-American-German Frontiers of Science (JAGFOS) Symposium

|                                                                           |      |
|---------------------------------------------------------------------------|------|
| <b>Towards Human-understandable Explanations with XAI 2.0</b>             | 2022 |
| AI4Good webinar series of the International Telecommunication Union (ITU) |      |
| <b>Towards Actionable XAI</b>                                             | 2022 |
| International Artificial Intelligence Doctoral Academy                    |      |

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## Supervision & Guidance

Collaboration with and (co-)supervision of 3 (Senior) Technical Staff, 2 PostDocs, 32 PhD Students, 40+ Master's Students, 3 Bachelor's Students and 9 Guest Researchers since 2017

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## Third-Party Funded Research Projects

9 third-party funded research projects acquired and managed since 2018, including

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|---------------------------------------------------------------------------|-------------|
| <b>xJuRAG</b>                                                             | 2025 - 2028 |
| Explainable AI für juristische RAG-Anwendungen                            |             |
| Funded with 1.2MM€ by BMBF                                                |             |
| <b>ACHILLES</b>                                                           | 2024 - 2028 |
| Human-Centred Machine Learning: Lighter, Clearer, Safer                   |             |
| Funded with 8.2MM€ by the European Union                                  |             |
| <b>TEMA</b>                                                               | 2022 - 2026 |
| Trusted Extremely Precise Mapping and Prediction for Emergency Management |             |
| Funded with 11.3MM€ by the European Union                                 |             |
| <b>iToBoS</b>                                                             | 2021 - 2025 |
| Intelligent Total Body Scanner                                            |             |
| Funded with 11.7MM€ by the European Union                                 |             |

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## Honors & Awards

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|---------------------------------------------------------------------------|-------------|
| <b>Digital Justice Award</b>                                              | 2025        |
| <b>Machine Learning and the Physical Sciences Reproducibility Badge</b>   | 2024        |
| <b>Stanford Top 2% Scientist Worldwide*</b>                               | 2022 - 2025 |
| <b>Best Short Paper Award @ ETRA</b>                                      | 2023        |
| <b>Pattern Recognition Best Paper Award and Pattern Recognition Medal</b> | 2020        |
| <b>Hugo-Geiger-Prize (1st place)</b>                                      | 2019        |
| <b>Freunde des HHI Nachwuchspreis</b>                                     | 2019        |
| <b>ERCIM Cor van Baayen Award (finalist)</b>                              | 2019        |
| <b>Best Paper Award @ ICMLW</b>                                           | 2016        |

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## Patents

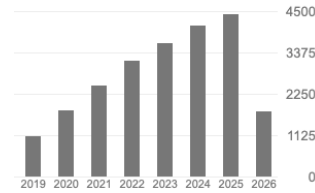
|                                                                                                 |      |
|-------------------------------------------------------------------------------------------------|------|
| <b>Determining a Solution of a Partial Differential Equation</b>                                | 2025 |
| <b>Output Token Selection in Machine Learning Models</b>                                        | 2025 |
| <b>Attention Head Functionalities in Machine Learning Models</b>                                | 2025 |
| <b>Relevance Score Assignment dealing with an Attention Module and Applications thereof</b>     | 2024 |
| <b>A Concept Representation of a Machine Learning Model</b>                                     | 2024 |
| <b>Analyzing an Inference of a Machine Learning Predictor</b>                                   | 2023 |
| <b>Method and System for Simulating an Optical Image of a Photonic and/or Electronic Device</b> | 2022 |
| <b>Pruning and/or Quantizing Machine Learning Predictors</b>                                    | 2020 |
| <b>Relevance Score Assignment for Artificial Neural Networks</b>                                | 2016 |

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## Publications

### Summary of Scientific Impact

|                | All   | Since 2021 |
|----------------|-------|------------|
| # Publications | 111   | 85         |
| # Citations    | 23803 | 19773      |
| h-index        | 40    | 39         |
| i10-index      | 68    | 66         |



per Google Scholar, retrieved on June 12<sup>th</sup>, 2026.

### Selected Publications

#### Theoretical & Methodological Contributions to XAI

1. Achibat, Hatefi, Dreyer, Jain, Wiegand, **Lapuschkin**, Samek (2024).  
“AttnLRP: Attention-Aware Layer-wise Relevance Propagation for Transformers”.  
In: *Proceedings of the 41st International Conference on Machine Learning (ICML)* 135–168.  
*In this paper we adapt the popular LRP method to contemporary Transformer architectures, yielding state-of-the-art explanation quality at exceptional computational efficiency. This achievement enables the analysis of the reasoning processes of Vision Transformers and LLMs in real time, eg. in chat bot applications.*
2. Achibat, Dreyer, Eisenbraun, Bosse, Wiegand, Samek and **Lapuschkin** (2023).  
“From attribution maps to human-understandable explanations through Concept Relevance Propagation”.  
In: *Nature Machine Intelligence* 5(9):1006–1019.  
*A paper introducing the second generation of Explainable Artificial Intelligence with human-readable and abstract concept-based explanations.*
3. Pahde, Dreyer, Samek and **Lapuschkin** (2023).  
“Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models”.  
In: *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention* 596–606.  
*This paper is dedicated to the incorporation of XAI as a standard component into the life cycle of Artificial Intelligence systems, with the intent to improve performance, reliability, and safety of AI.*
4. Hedström, Weber, Krakowczyk, Bareeva, Motzkus, Samek, **Lapuschkin** and Höhne (2023).  
“Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond”.  
In: *Journal of Machine Learning Research* 24(34):1–11.  
*In this paper we present the Quantus toolkit, the first-ever comprehensive XAI evaluation toolkit, constituting a well-organized collection of metrics and tutorials for evaluating explainable models, driven by community contributions.*
5. Montavon, **Lapuschkin**, Binder, Samek and Müller (2017).  
“Explaining NonLinear Classification Decisions with Deep Taylor Decomposition”.  
In: *Pattern Recognition* 65:211–222.  
*A paper discussing the mathematical foundation of LRP and its properties. Pattern Recognition Best Paper Award and Pattern Recognition Medal winner of 2020.*
6. **Bach**, Binder, Montavon, Klauschen, Müller and Samek (2015).  
“On Pixel-wise Explanations for Non-Linear Classifier Decisions by Layer-wise Relevance Propagation”.  
In: *PLoS ONE* 10(7):e0130140.  
*A very influential and early work on local XAI, introducing the widely used Layer-wise Relevance Propagation method. This work has so far received over 6800 citations as counted by Google Scholar.*

#### Applications of XAI

7. Kahardipraja, Achibat, Wiegand, Samek and **Lapuschkin** (2025).  
“The Atlas of In-Context Learning: How Attention Heads Shape In-Context Retrieval Augmentation”.  
In: *Advances in Neural Information Processing Systems (NeurIPS)* .  
*In this work we use our group’s state-of-the-art XAI techniques in order to differentiate between different types of attention heads and their roles in LLMs. We use the gained insights in order to obtain reproducible control over the generation process via the isolation of function vectors. We further gain the capability predict whether a candidate*

token ranked highly for prediction is grounded in context or in parametric knowledge, allowing users to select a preferred generation path. With our technique also obtain hallucination-free and near-causal citation capabilities for each predicted token at zero additional computational cost, solving a fundamental problem with safe and reliable deployment of LLMs.

8. Anders, Weber, Neumann, Samek, Müller and **Lapuschkin** (2022).  
“Finding and Removing Clever Hans: Using Explanation Methods to Debug and Improve Deep Models”.  
In: *Information Fusion* 77:261–295.  
*The authors’ first work in a series dedicated to the exploitation of knowledge derived from XAI for the improvement of performance and robustness of AI systems.*
9. Yeom, Seegerer, **Lapuschkin**, Binder, Wiedemann, Müller and Samek (2021).  
“Pruning by Explaining: A Novel Criterion for Deep Neural Network Pruning”.  
In: *Pattern Recognition* 115:107899.  
*In this paper we leverage information about the importance of latent neural network structures obtained through XAI, in order to drastically reduce over-parameterization by identifying and removing non-critical components, resulting in up to 95% smaller models without loss of performance, and thus strong gains in energy and run time efficiency.*
10. **Lapuschkin**, Wäldchen, Binder, Montavon, Samek and Müller (2019).  
“Unmasking Clever Hans Predictors and Assessing what Machines Really Learn”.  
In: *Nature Communications* 10:1069.  
*One of the first papers to rigorously perform model- and data analysis through the lens of XAI, adding a voice of caution to the ongoing excitement about machine intelligence.*